

**PROJECT REPORT**

**AI E-COMMERCE CUSTOMER SUPPORT  
AGENT**

*A Project Report Submitted in Partial Fulfilment of the Requirements for the Academic Project*

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# 1. PROJECT OVERVIEW

## 1.1 Project Title

AI E-Commerce Customer Support Agent

## 1.2 Problem Statement

E-commerce customers frequently need quick assistance with product questions, order-status requests, returns, and product recommendations. Handling these requests manually can lead to delays, inconsistent responses, and repeated support work. The proposed AI E-Commerce Customer Support Agent is designed to provide a conversational support interface that can understand customer requests and use appropriate tools and stored context to respond efficiently.

## 1.3 Brief Description of the Project

The project develops an Agentic AI customer-support assistant for an e-commerce environment. The agent handles product-related queries, checks order status, assists with return-related requests, and provides recommendations. The use-case specification identifies Tool Calling + Memory as the key agent capabilities. Tool calling enables the agent to interact with external functions such as order lookup or product information services, while memory helps maintain useful conversational context.

## 1.4 Scope

- Answer common product and shopping-related customer questions.
- Retrieve order information through an appropriate tool.
- Support return-related conversations and guide customers through the process.
- Provide product recommendations based on the customer's stated needs.
- Maintain relevant conversational context using memory.

## 2. OBJECTIVES & PROPOSED SOLUTION

### 2.1 Project Objectives

- Build an AI-powered customer support assistant for common e-commerce requests.
- Use tool calling to obtain information that is not available from the conversation alone.
- Use memory to preserve relevant context across a customer interaction.
- Provide clear, conversational responses for product, order, return, and recommendation requests.
- Reduce repetitive support effort by automating common customer-service interactions.

### 2.2 How the Agentic AI Solution Works

The assistant follows an agentic workflow. First, it receives the customer's message and interprets the intent. If the request can be answered directly, the agent responds using its available knowledge. If external information is required, the agent selects an appropriate tool, passes the required input, receives the tool result, and incorporates that result into the response. Relevant conversational information can be retained through memory so that later turns remain consistent with the ongoing interaction.

### 2.3 High-Level Workflow

| Stage                   | Function                                                                              |
|-------------------------|---------------------------------------------------------------------------------------|
| 1. Customer Query       | Customer enters a product, order, return, or recommendation request.                  |
| 2. Intent Understanding | Agent interprets the request and determines the required action.                      |
| 3. Memory Check         | Relevant previous conversation context is considered.                                 |
| 4. Tool Selection       | Agent decides whether an external tool is needed.                                     |
| 5. Tool Execution       | Selected tool returns the required e-commerce information.                            |
| 6. Response Generation  | Agent combines the available context and tool result into a customer-friendly answer. |

Agentic Workflow — AI E-Commerce Customer Support Agent



Key capabilities: Tool Calling (product / order / return functions) + Memory (conversational context retention)

Figure 1 — Agentic workflow: the six stages from customer query to response generation.

### 2.4 Key Features

- Product Query Handling: Responds to questions about products and shopping needs.

- Order Status: Uses tool calling to retrieve order-related information when required.
- Returns Support: Guides customers through return-related requests.
- Recommendations: Suggests relevant products based on customer requirements.
- Memory: Maintains useful conversational context during the interaction.

## 3. IMPLEMENTATION & RESULTS

### 3.1 Technologies / Tools Used

The provided use-case specification identifies Agentic AI capabilities rather than a fixed technology stack. Therefore, the implementation can be organized around an AI agent, tool-calling functions, and a memory component. The exact programming language, framework, model provider, database, and deployment platform are not specified in the supplied source.

| Component                  | Purpose                                                       |
|----------------------------|---------------------------------------------------------------|
| AI Agent                   | Understands customer intent and decides the next action.      |
| Tool Calling               | Connects the agent to order/product/return-related functions. |
| Memory                     | Retains relevant conversational context.                      |
| E-commerce Data / Services | Provides information required for customer-support tasks.     |

**System Architecture — Component View**

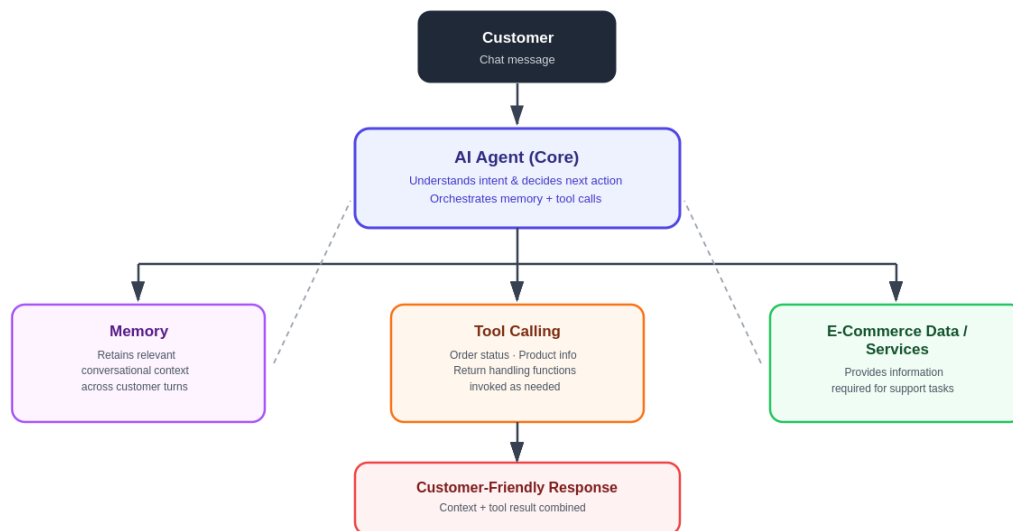


Figure 2 — Component-level system architecture.

### 3.2 Working Process

A typical interaction begins when a customer enters a request such as asking about a product, checking an order, requesting a return, or seeking a recommendation. The agent interprets the request, checks relevant memory, and decides whether a tool call is necessary. After receiving the required information, the agent produces a concise response and preserves useful context for subsequent turns.

### 3.3 Example Interaction Flows

| Customer Request              | Agent Action                                                      | Expected Response                                                   |
|-------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|
| “Where is my order?”          | Identify order-status intent → call order-status tool.            | Provide the returned order-status information.                      |
| “I want to return this item.” | Identify return intent → guide or invoke return-related function. | Explain the applicable return steps based on available information. |
| “Suggest a product for me.”   | Use stated preferences and relevant context.                      | Provide suitable product recommendations.                           |
| “Tell me about this product.” | Interpret product query and use available product information.    | Give a clear product-focused answer.                                |

### 3.4 Screenshots / Output

The supplied project brief did not include screenshots of an implemented application. To make the intended interface concrete, the mockups below illustrate how each output area is meant to look and behave; they are UI mockups created for this report, not screenshots of a running system.

| Output Area    | Demonstration to Capture                                                     |
|----------------|------------------------------------------------------------------------------|
| Chat Interface | Customer enters a support question and receives an agent response.           |
| Order Lookup   | Agent invokes the order-status tool and displays the returned status.        |
| Return Support | Agent guides the customer through a return-related request.                  |
| Recommendation | Agent produces a product recommendation from the customer's requirements.    |
| Memory         | A follow-up question demonstrates that relevant earlier context is retained. |

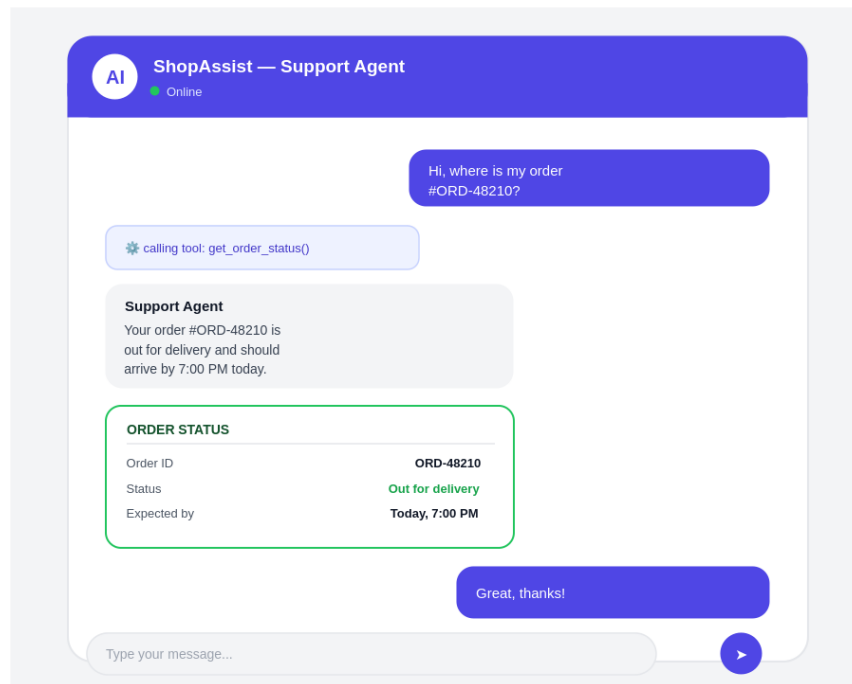


Figure 3 — Chat interface: order-status lookup with tool-call indicator, response, and status card.



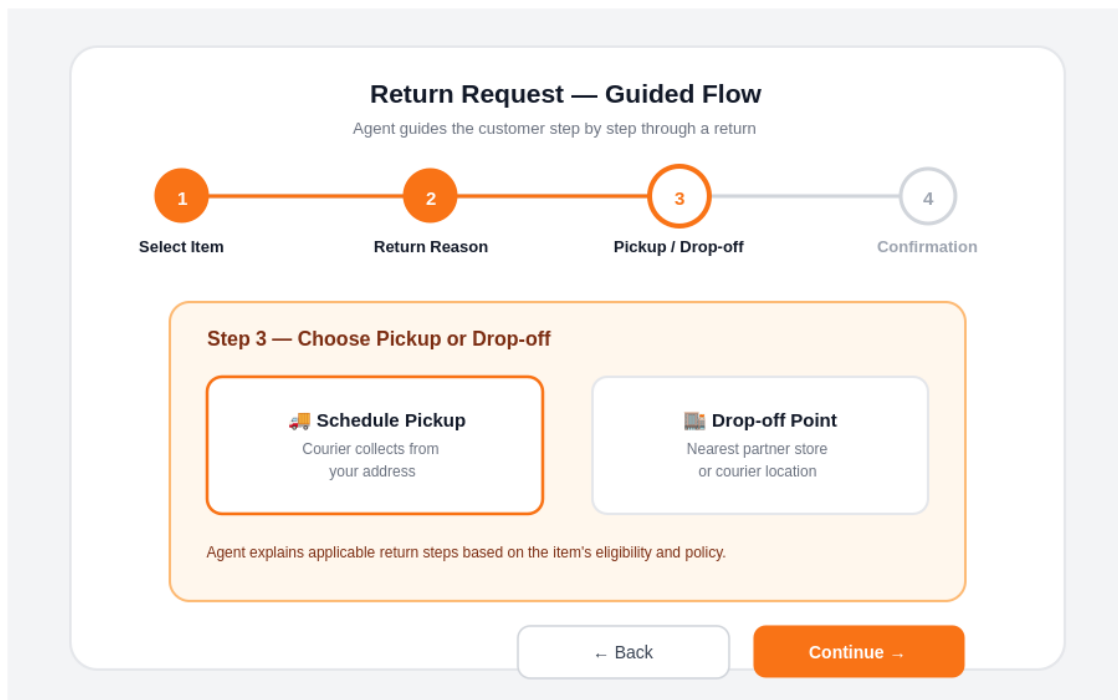


Figure 4 — Guided, step-by-step return flow.

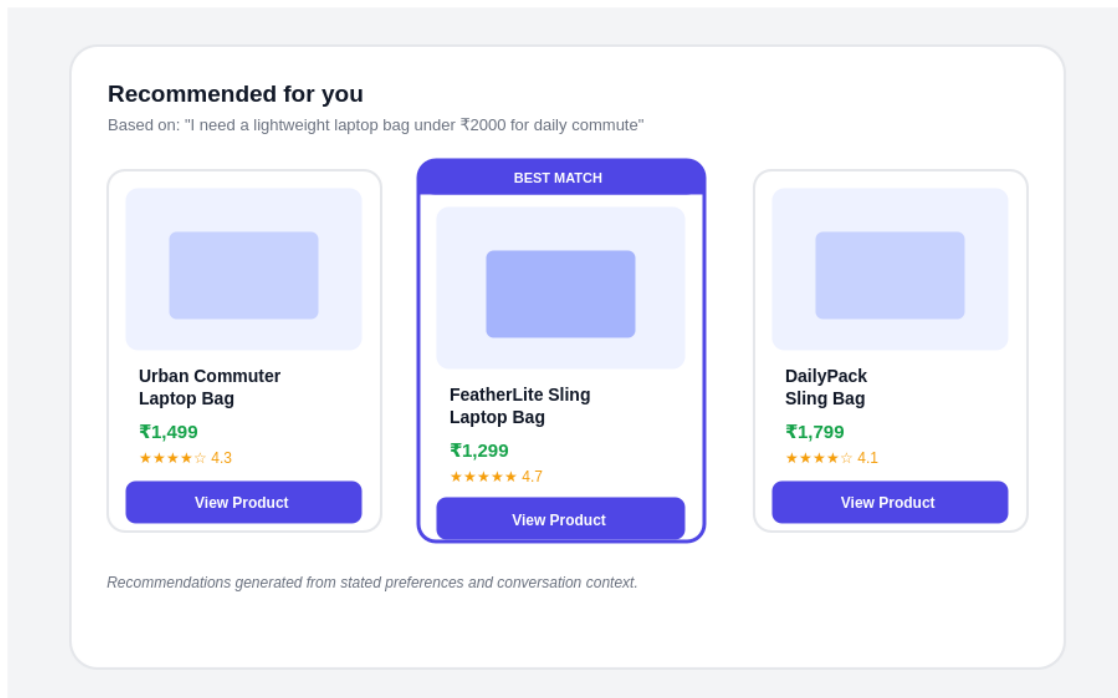


Figure 5 — Product recommendation cards generated from stated customer preferences.

### 3.5 Results Achieved / Expected Demonstration

Based on the supplied use-case description, the intended result is an e-commerce customer-support agent capable of handling product queries, order-status requests, returns, and recommendations, with Tool Calling + Memory as its key agent capabilities. Actual measured accuracy, response time, user-satisfaction scores, or production results are not provided in the source material and are therefore not claimed in this report.

- Product-related customer questions can be handled through a conversational agent.
- Order-status requests can be connected to a dedicated tool.
- Return-related requests can be supported through the agent workflow.
- Recommendations can be generated from customer requirements and conversation context.
- Memory can preserve relevant information across the interaction.

## 4. CONCLUSION & FUTURE SCOPE

### 4.1 Project Conclusion

The AI E-Commerce Customer Support Agent provides a practical Agentic AI approach to common e-commerce support activities. Instead of limiting the system to static question answering, the agent can decide when to use tools and can retain useful conversation context through memory. The proposed workflow covers the four main functions identified in the selected use case: product queries, order status, returns, and recommendations.

### 4.2 Challenges Faced

- Determining when a customer request requires an external tool.
- Maintaining consistent context across multiple turns.
- Handling incomplete or ambiguous customer requests.
- Ensuring that tool results are presented clearly and accurately.
- Connecting the conversational agent with reliable e-commerce data sources.

### 4.3 Future Enhancements

- Add authentication so customers can securely access their own order information.
- Integrate live inventory, delivery tracking, and payment-related services.
- Add multilingual customer support.
- Introduce human-agent handoff for complex or exceptional cases.
- Add analytics for common support intents and customer-service performance.
- Improve recommendation quality using customer preferences and purchase history.
- Add stronger validation and error handling around tool calls.

## 5. REFERENCES

1. Project Template for IBM Internship – Agentic AI, supplied project document. The document specifies the report structure: Project Overview; Objectives & Proposed Solution; Implementation & Results; Conclusion & Future Scope; and References.
2. Selected Use Case: AI E-Commerce Customer Support Agent. The supplied use-case slide describes the project as handling product queries, order-status requests, returns, and recommendations, with Tool Calling + Memory as key agent capabilities.

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